

# Equilibrium

Ingyu Bahng

May 2026

## Contents

|   |                             |   |
|---|-----------------------------|---|
| 1 | Reaction Rate & Equilibrium | 2 |
| 2 | Types of Equilibria         | 5 |
| 3 | Shifting Equilibria         | 7 |

# 1 Reaction Rate & Equilibrium

## Core Concept 1.1: Reaction Rates

Atoms, ions, and molecules react with one another via collisions. The more frequent and energetic the collisions, the faster the reaction rate.

Take the formation of compound C as an example:  $A + B \longrightarrow C$ . The rate at which C forms depends on factors such as  $[A]$ ,  $[B]$ , temperature, the presence of a catalyst, etc. and can be expressed mathematically as:

$$\text{Rate} = k[A]^m[B]^n$$

Here,  $k$  is the **rate constant**, which is specific to the given reaction and its conditions.  $m$  and  $n$  are the **partial reaction orders**, which represents a reactant's influence over the reaction rate. Here is a hypothetical plot of  $[A]$ ,  $[B]$ , and  $[C]$  over time:

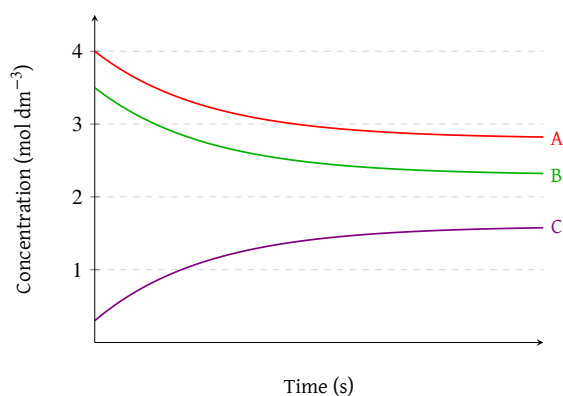


Figure 1: Equilibrium of  $A + B \longrightarrow c$

In single-step reactions, partial reaction orders are equal to the stoichiometric coefficients of in the balanced chemical equation. However, in multi-step reactions, partial reaction orders may not directly correspond to the stoichiometric coefficients and must be determined experimentally.

Consider the decomposition of  $H_2O_2$  ( $H_2O_2 \longrightarrow H_2O + O_2$ ) as an example. When you graph the concentration of  $H_2O_2$  over time, you get an exponential decay curve, where the decomposition rate of  $H_2O_2$  decreases as  $[H_2O_2]$  decreases.

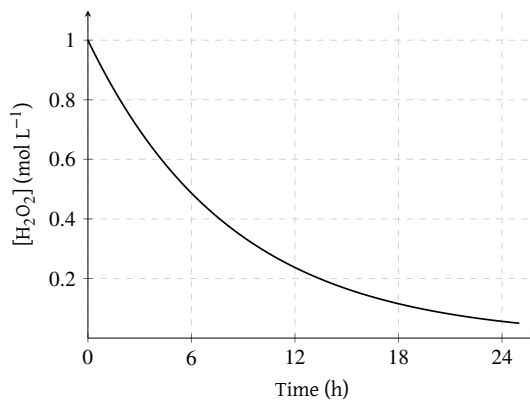


Figure 2: Decomposition of  $H_2O_2$  over time

**Core Concept 1.2: Equilibrium**

Reactions do not always proceed in only one direction. Many reactions are **reversible**, meaning that the products can react to reform the reactants. When the rates of the forward and reverse reactions are equal, the system is said to be at **equilibrium**. Thus, for the general reversible shell reaction  $AB \rightleftharpoons A + B$ , the Eq reaction state can be expressed as:

$$k_{forward}[AB] = k_{backward}[A][B]$$

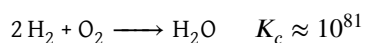
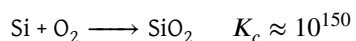
And the equilibrium constant  $K_c$  is defined as:

$$K_c = \frac{k_f}{k_b} = \frac{[A][B]}{[AB]}$$

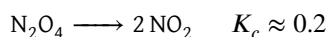
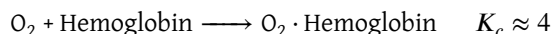
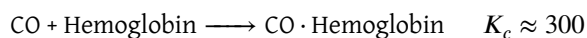
**Core Concept 1.3: Analyzing  $K_c$** 

$K_c$  provides insight into the position of equilibrium. A large  $K_c$  ( $\gg 1$ ) indicates product favorability, while a small  $K_c$  ( $\ll 1$ ) indicates reactant favorability.

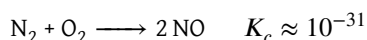
Some examples of reactions that possess a **large**  $K_c$  (reaction to completion) include:



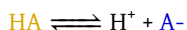
Some examples of reactions that possess a **moderate**  $K_c$  (half-reaction) include:



An example of a reaction that possess a **negligible**  $K_c$  (no reaction) include:



The decomposition of Bromothymol Blue (denoted as A) exhibits a nice visual representation of reaction equilibria.



If we start with a pure HA solution, the solution will gradually change color from **yellow** to **green** to **blue** as the forward reaction proceeds and  $[\text{A}^-]$  rises. This particular compound is also known as an indicator, a concept covered in ??.

**Core Concept 1.4: Le Chatelier's Principle**

**Le Chatelier's Principle** states that a system in equilibrium subjected to a stress will react in a direction that tends to reduce that stress and return back to Eq.

In more exact terms, stress will shift the **reaction quotient (Q)** — which represents the relative amounts of reactant and products at a given point in time before it has necessarily reached equilibrium ( $Q \neq K$ ) — and the system will react in a manner that shifts Q back to K.

$$Q = \frac{[\text{A}]_{\text{current}}[\text{B}]_{\text{current}}}{[\text{AB}]_{\text{current}}}$$

There are three major categories of stress in the context of Le Chatelier's Principle.

### Core Concept 1.5: Le Chatelier's Principle (Concentration Change)

Shifting the point concentrations of any chemical participant will shift  $Q$ .

Let's reexamine the Bromothymol Blue reaction ( $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ ). If we add more  $\text{H}^+$  to an initially green solution ( $[\text{HA}] = [\text{A}^-]$ ), the system will increase the backwards reaction rate to consume excess  $\text{H}^+$ , resulting in a more yellow solution. Conversely, removing  $\text{H}^+$  (by adding  $\text{OH}^-$ ) will increase the forward reaction rate to produce more  $\text{H}^+$ , resulting in a more blue solution.

Consider another example ( $\text{PCl}_5 \longrightarrow \text{PCl}_3 + \text{Cl}_2$ ) based on the visual below. The plot shows the abrupt addition of  $\text{PCl}_5$  to a system originally at equilibrium. Consequently,  $Q < K$  and the system reacts by increasing the forwards reaction rate to re-establish equilibrium, resulting in a subsequent increase in  $[\text{PCl}_3]$  and  $[\text{Cl}_2]$ .

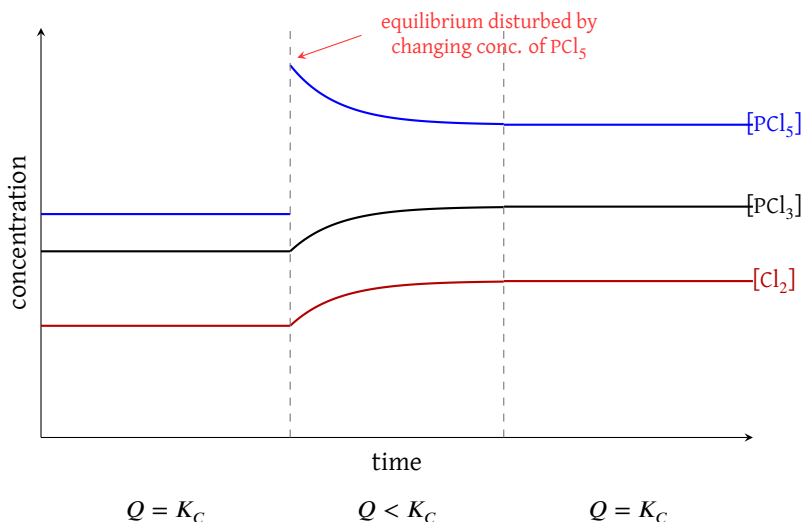


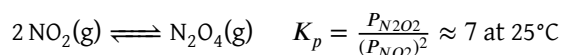
Figure 3: Disturbance of equilibrium by changing concentration of  $\text{PCl}_5$

### Core Concept 1.6: Le Chatelier's Principle (Pressure & Volume Change)

While concentration is the primary focus for solutions, pressure and volume are the key variables for gaseous systems. Increasing the pressure forces the equilibrium toward the side with fewer gas molecules to relieve the stress. Conversely, decreasing pressure allows the reaction to shift toward the side with more gas molecules.

The equilibrium constant in terms of partial pressures for gaseous reactions is written as  $K_p$ .

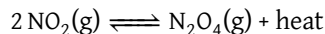
Consider the following reaction:



Increasing the volume ( $V$ ) causes a drop in total pressure ( $P$ ). To counteract this, the equilibrium shifts to the left, favoring the side with more moles of gas. Mathematically, this occurs because the partial pressure of  $\text{NO}_2$  is squared in the denominator of the  $Q_p$  expression. A decrease in pressure impacts the squared term more significantly than the linear numerator, making  $Q < K$  and forcing a shift toward the reactants.

**Core Concept 1.7: Le Chatelier's Principle (Temperature Change)**

A temperature shift is analogous to an increase or decrease of heat within a system, and the Q shift direction depends on whether heat is a net product (**exothermic**) or a reactant (**endothermic**) of the chemical reaction at hand. Reconsider the following gaseous reaction in the context of heat.



Much like the addition of  $\text{N}_2\text{O}_4$ , increasing the temperature drives the equilibrium to the left to consume the excess energy. Conversely, cooling the system shifts the reaction to the right as it attempts to regenerate the lost heat.

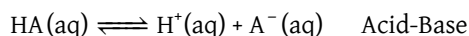
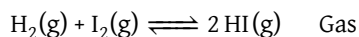
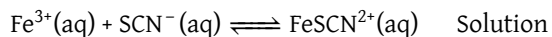
Chemical equilibrium in the context of temperature is elaborated more upon in Section 3.

## 2 Types of Equilibria

Equilibria can also be categorized into types.

**Core Concept 2.1: Homogeneous Equilibria**

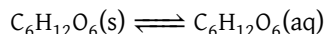
In **homogeneous equilibria**, all reactants and products are in the same phase (solid, liquid, or gas), such as the following examples.



In **heterogeneous equilibria**, the reactants and products are in different phases and can be further subcategorized into three domains.

**Core Concept 2.2: Heterogeneous Equilibria (Solubility of Solids)**

The solubility of solids equilibrium refers to the dissociation and crystallization rate of solutes. Consider the dissolution of glucose. Solutes that do not dissociate are called **molecular or nonelectrolyte solutes**.



$$\text{Dissociation Rate} = k_f(SA)$$

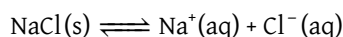
$$\text{Crystallization Rate} = k_c(SA)[\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})]$$

The dissociation rate only depends on the surface area (SA) of the solid glucose, while the crystallization rate depends on both the SA and  $[\text{C}_6\text{H}_{12}\text{O}_6]$ . At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})]$$

For glucose, the  $K_{sp}$  value is 5.1M at 25°C, meaning that the maximum  $[\text{C}_6\text{H}_{12}\text{O}_6]$  in water is 5.1M. Beyond this concentration, any excess glucose will remain undissolved.

Now consider a different compound like NaCl. Solutes that dissociate are called **ionic or electrolyte solutes**.



$$\text{Dissociation Rate} = k_f(SA)$$

$$\text{Crystallization Rate} = k_c(SA)[\text{Na}^+(\text{aq})][\text{Cl}^-(\text{aq})]$$

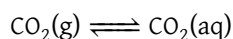
At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [\text{Na}^+(\text{aq})][\text{Cl}^-(\text{aq})]$$

At 25°C, the  $K_{sp} \approx 37.3$ , making the maximum  $[\text{NaCl}]$  in water approximately 6.11 M.

### Core Concept 2.3: Heterogeneous Equilibria (Solubility of Gases) & Henry's Law

Gases can dissolve into liquids just as how solids can. Consider the  $K_{sp}$  of  $\text{CO}_2$  in water.



When we reach equilibrium, we have the following:

$$K_H = \frac{[\text{CO}_2(\text{aq})]}{P_{\text{CO}_2}} \rightarrow [\text{CO}_2(\text{g})] = K_H P_{\text{CO}_2}$$

The equilibrium formula above is known as **Henry's Law**, and the constant  $K_H$  is called the **Henry's Constant** ( $M/atm$ ), a quantitative measure of the proportion between a gas's solubility in a liquid and its partial pressure. The higher the  $K_H$  value, the easier it is for a gas to dissolve in a liquid.

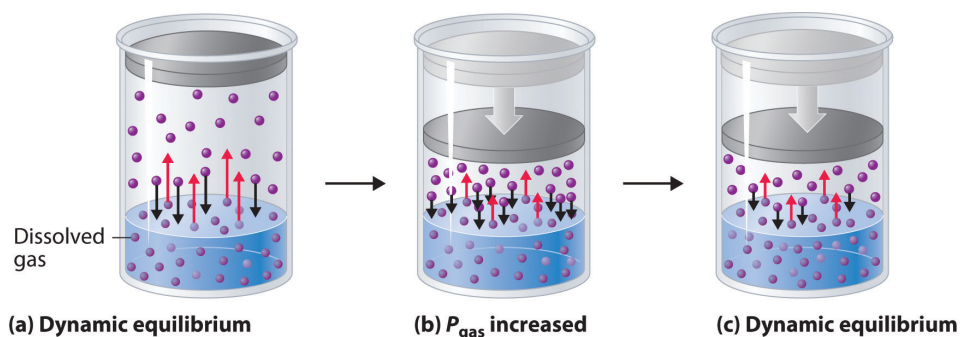
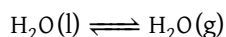


Figure 4: Gas solubility dynamic equilibrium

### Core Concept 2.4: Heterogeneous Equilibria (Phase Change)

Phase changes also experience equilibria. Consider the vaporization of water in a sealed flask.



$$\text{Evaporation Rate} = k_f(SA)$$

$$\text{Condensation Rate} = k_c(SA) P_{\text{H}_2\text{O}}$$

Just as the solubility equilibria for nonelectrolyte solutes is only dependent on the solvent concentration, the phase change equilibria for a gas to liquid transition is only dependent on  $P_{\text{H}_2\text{O}}$ .

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = P_{\text{H}_2\text{O}}$$

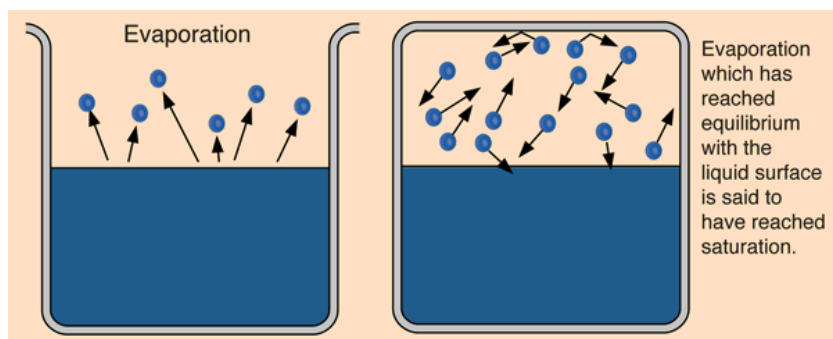


Figure 5: Phase change equilibrium of liquid-gas systems

*Phase change equilibria become irrelevant for solid-liquid transitions because both pure solids and pure liquids have fixed, constant concentrations that are excluded from the equilibrium expression, leaving no variable quantities and rendering the equilibrium constant meaningless.*

### 3 Shifting Equilibria

In **phase change equilibria**, the melting and boiling points of  $\text{H}_2\text{O}$  at  $0^\circ\text{C}$  and  $100^\circ\text{C}$  respectively are defined at a standard pressure of 1 atm; under varying pressure conditions, these transition temperatures become dynamic.

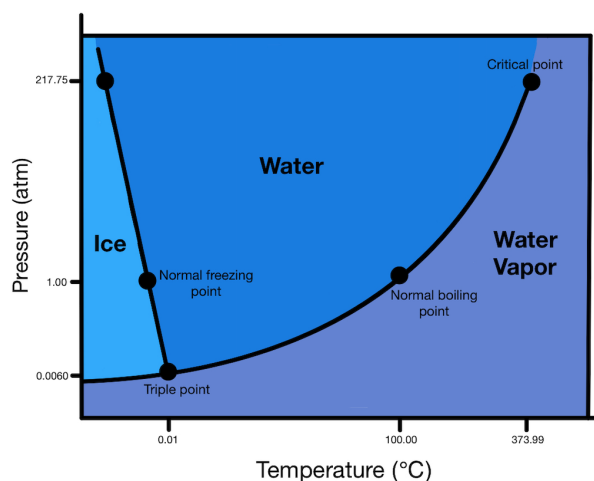


Figure 6: Phase diagram of water

Temperature shifts have a more nuanced effect in the context of equilibrium compared to the simple Le Chatelier's principle we have seen in Core Concept 1.7. Namely, the  $K_{sp}$  values of various solutes change across temperature ranges.

On the molecular scale,  $K_{sp}$  values of solid solutes increase with higher temperatures because the increased molecular motion of the solute and solvent molecules raises the probability of the solute entering the aqueous state.

In the context of gas solubility within a closed system, the Henry's Law constant  $K_H$  decreases with increasing temperature. Since gaseous molecules possess greater kinetic energy than their aqueous counterparts, the dissolution of a gas is an exothermic process; as temperature rises, the equilibrium shifts to favour the gaseous state, increasing partial pressure  $P$  at the expense of aqueous concentration  $C$ , thereby reducing  $K_H = \frac{C}{P}$ .

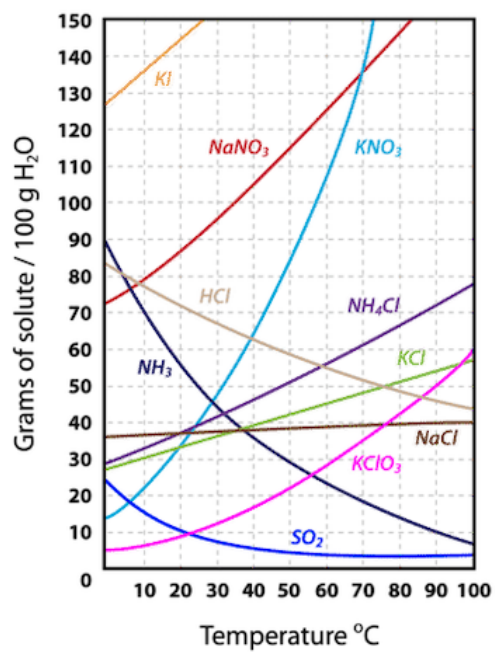


Figure 7: Solubility curve of various compounds